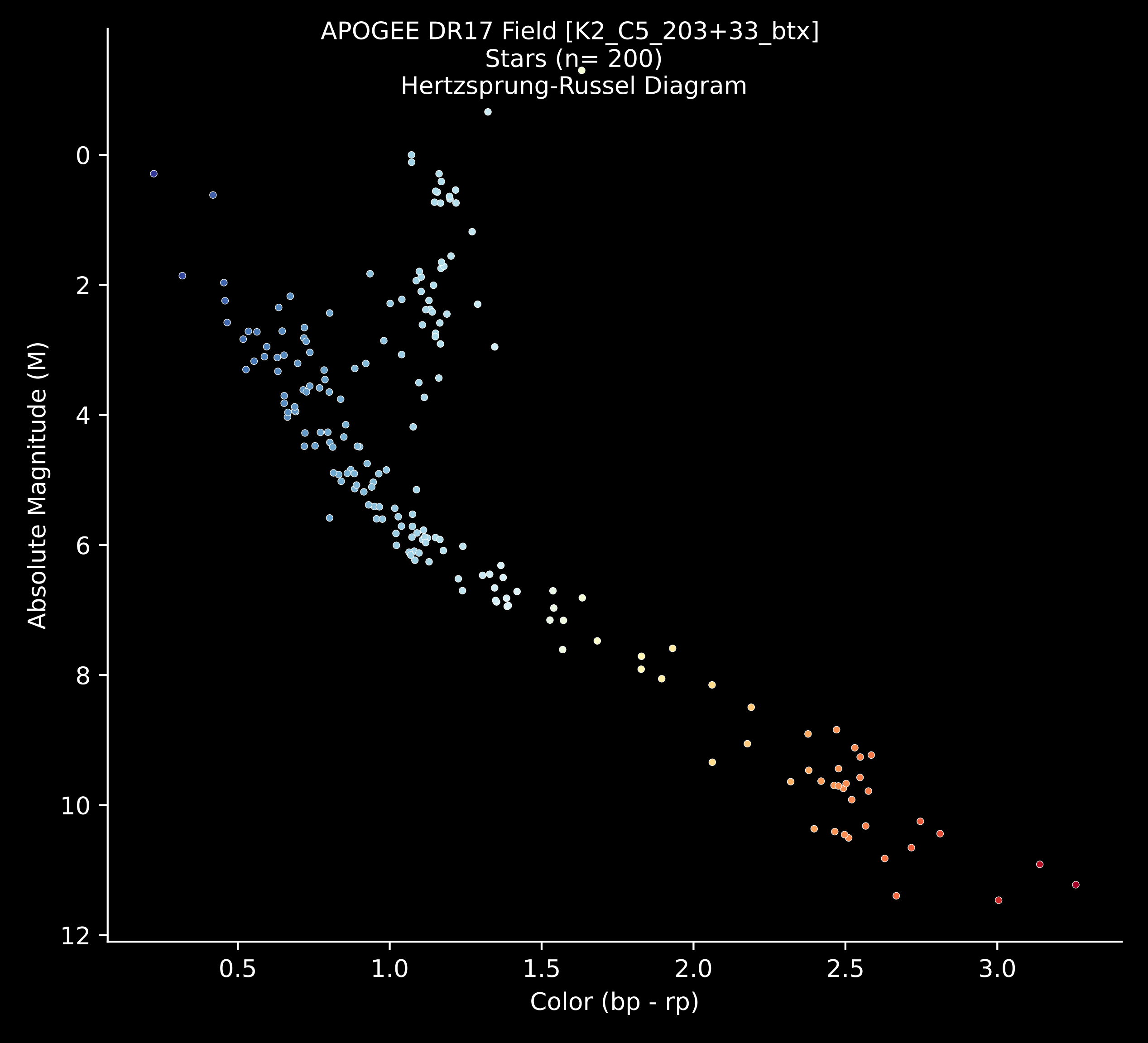
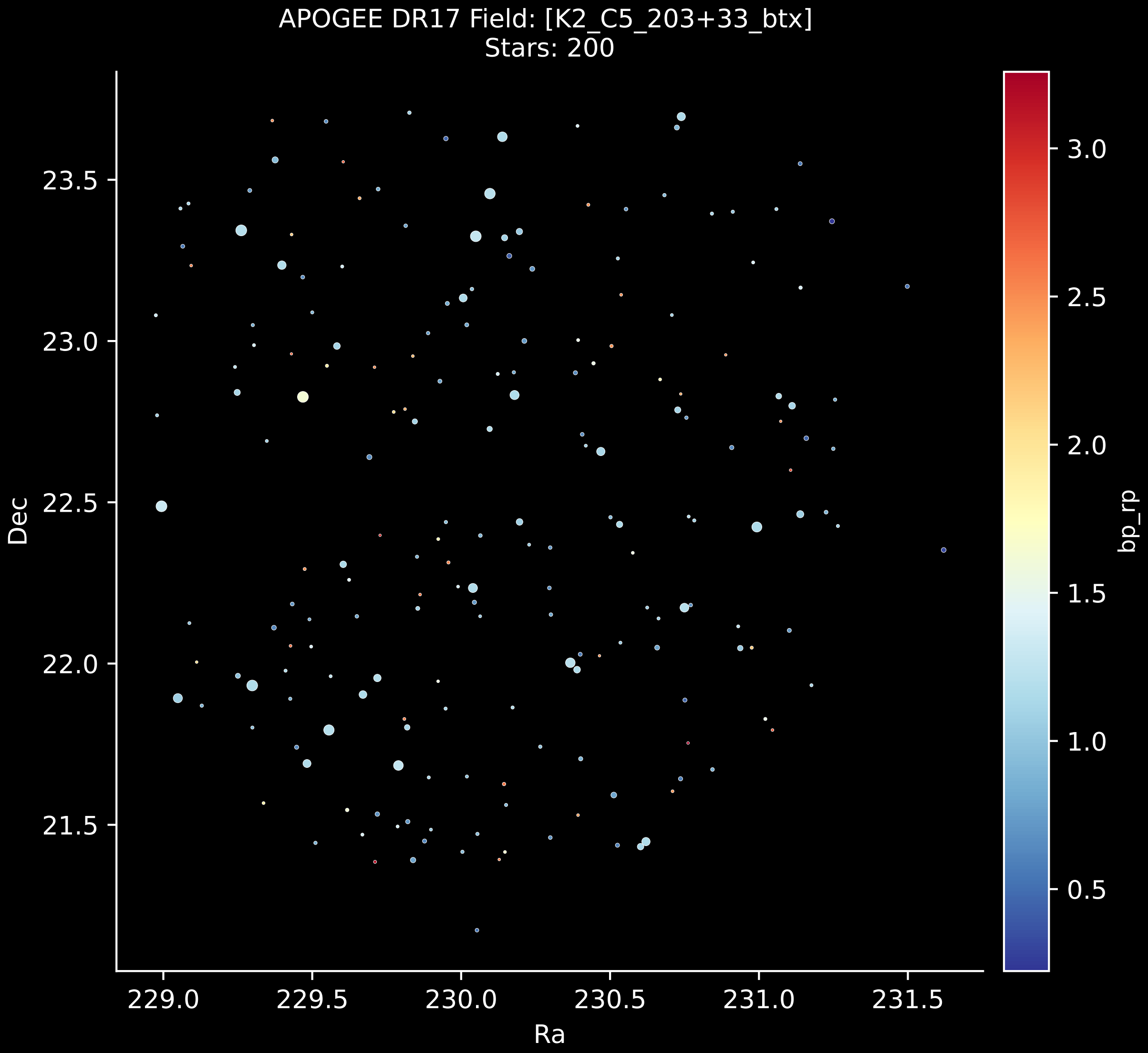
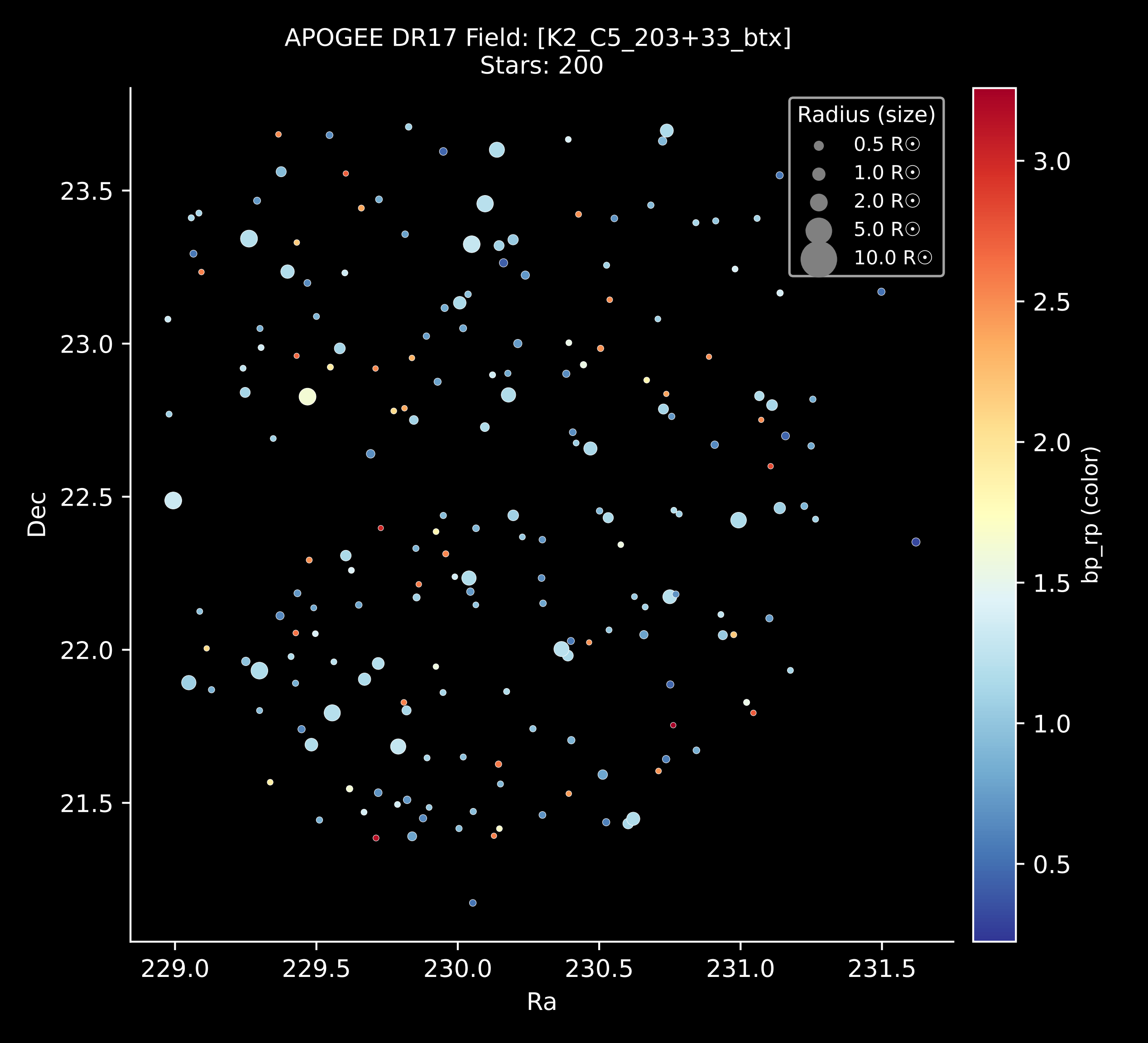
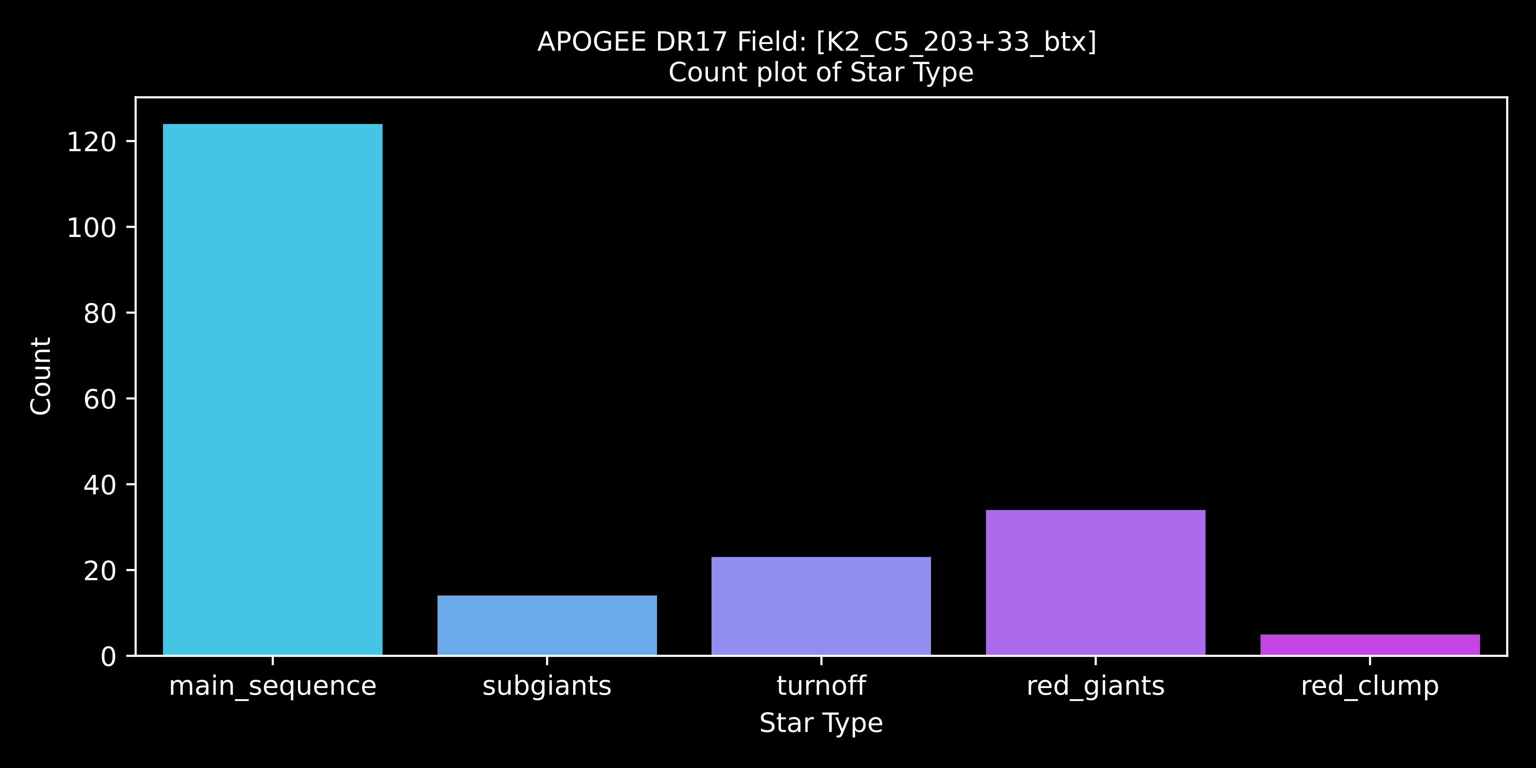
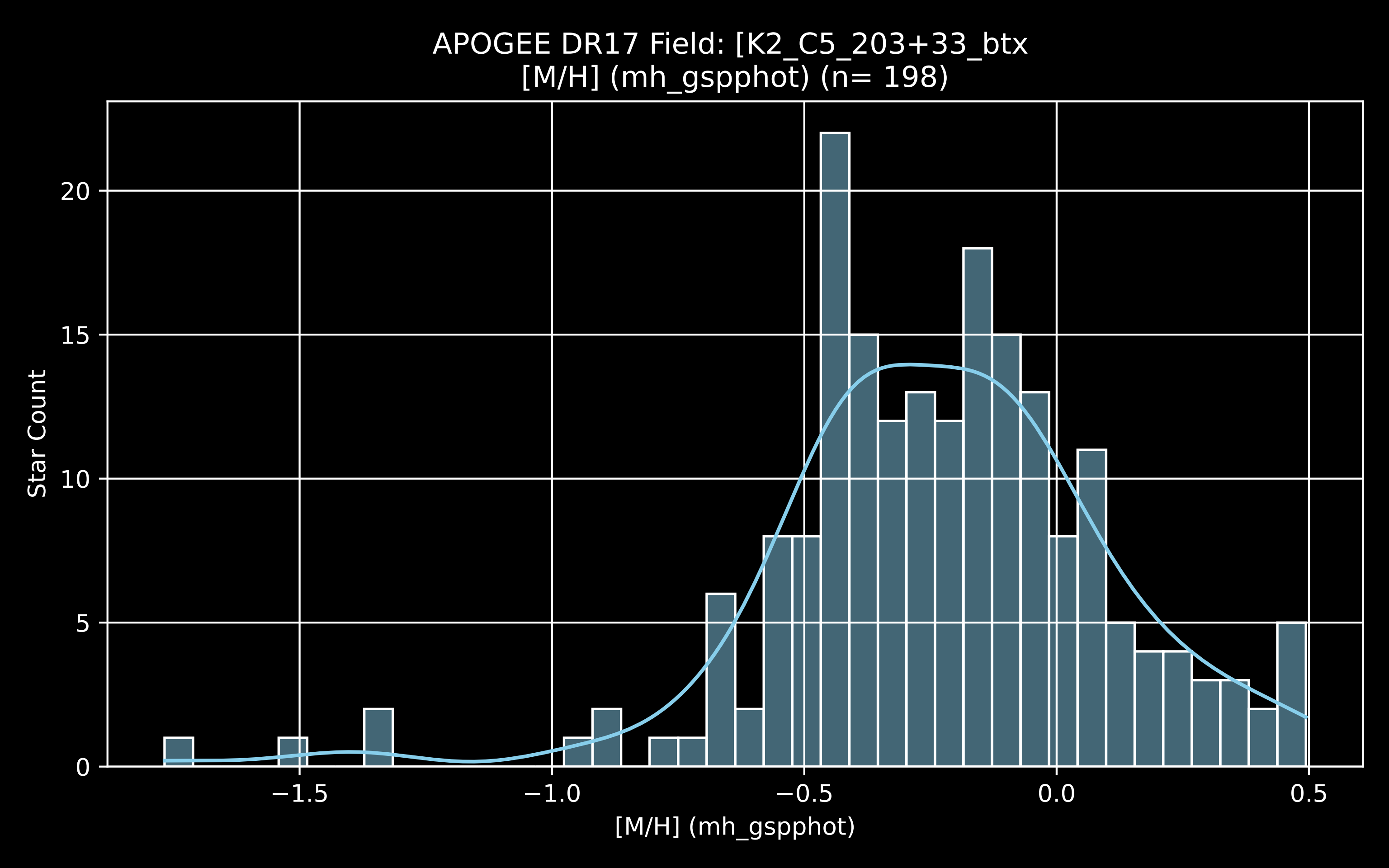


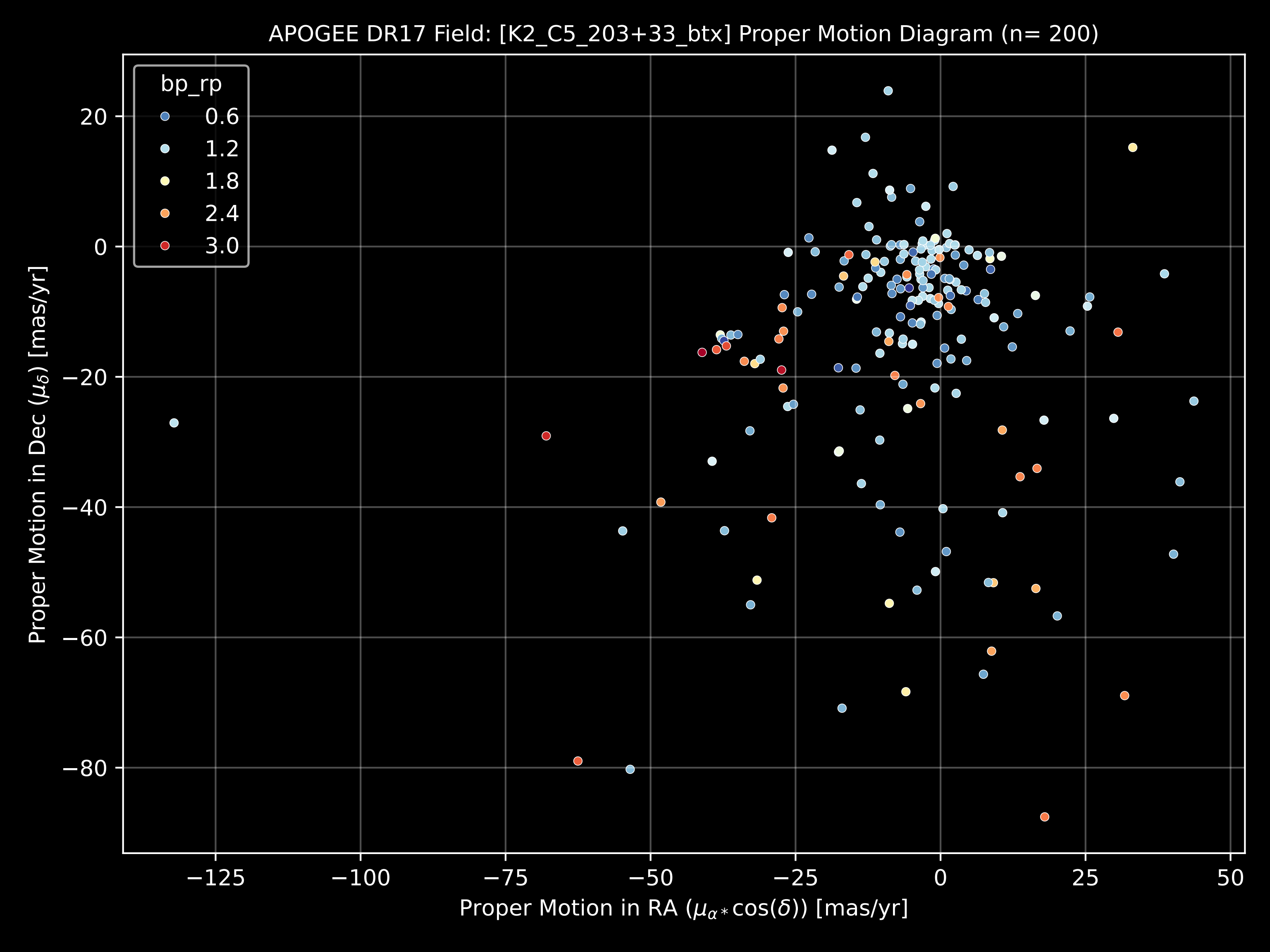
APOGEE DR17 Field [K2_C5_203+33_btx]
Stars (n= 200)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
0
2
4
6
8
10
12
0.5
1.0
1.5
2.0
2.5
3.0

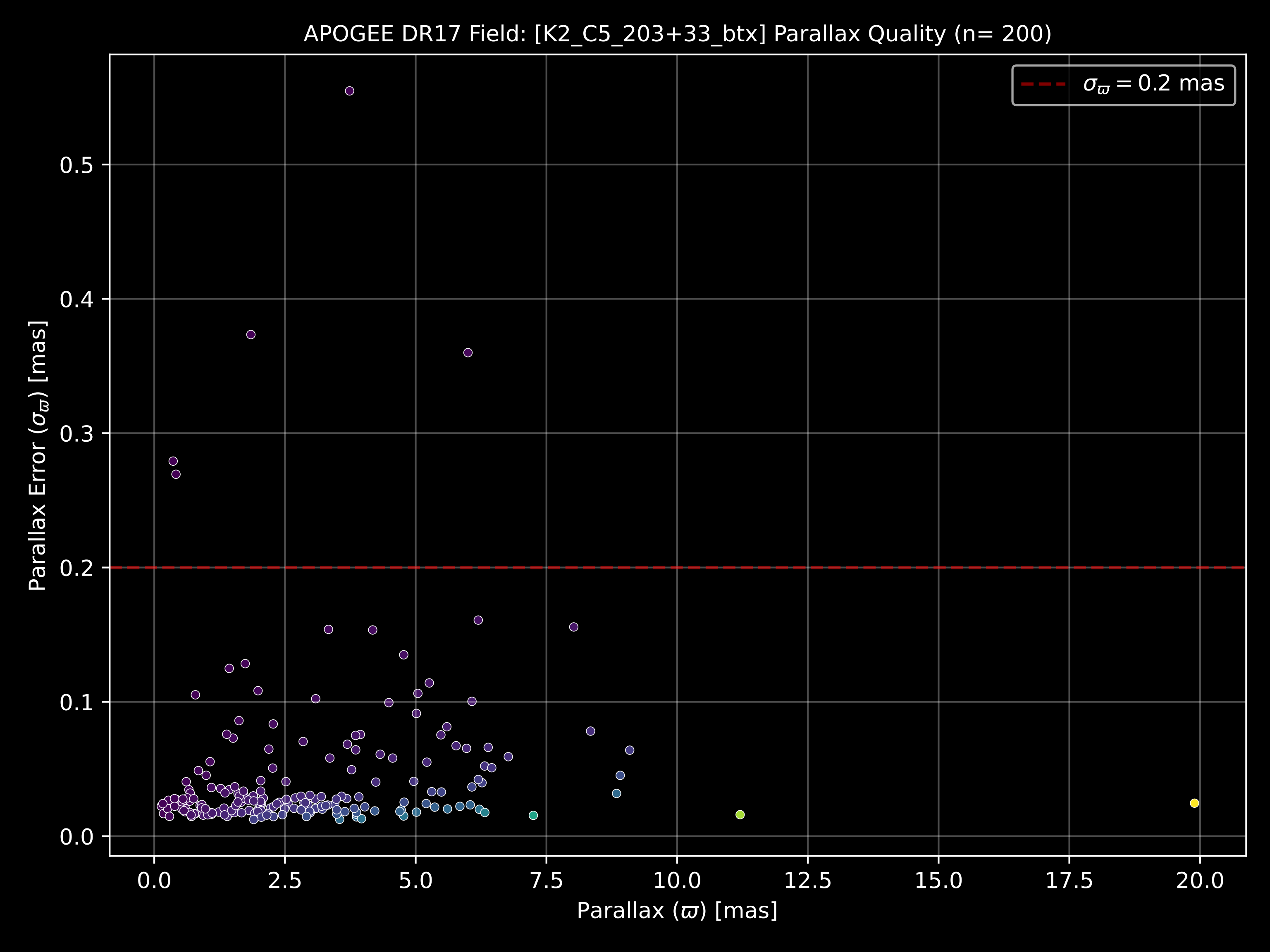

APOGEE DR17 Field: [K2_C5_203+33_btx]
Stars: 200
Dec
Ra
bp_rp
23.5
23.0
22.5
22.0
21.5
229.0
229.5
230.0
230.5
231.0
231.5
3.0
2.5
2.0
1.5
1.0
0.5


APOGEE DR17 Field: [K2_C5_203+33_btx]
Stars: 200
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
23.5
23.0
22.5
22.0
21.5
229.0
229.5
230.0
230.5
231.0
231.5
3.0
2.5
2.0
1.5
1.0
0.5

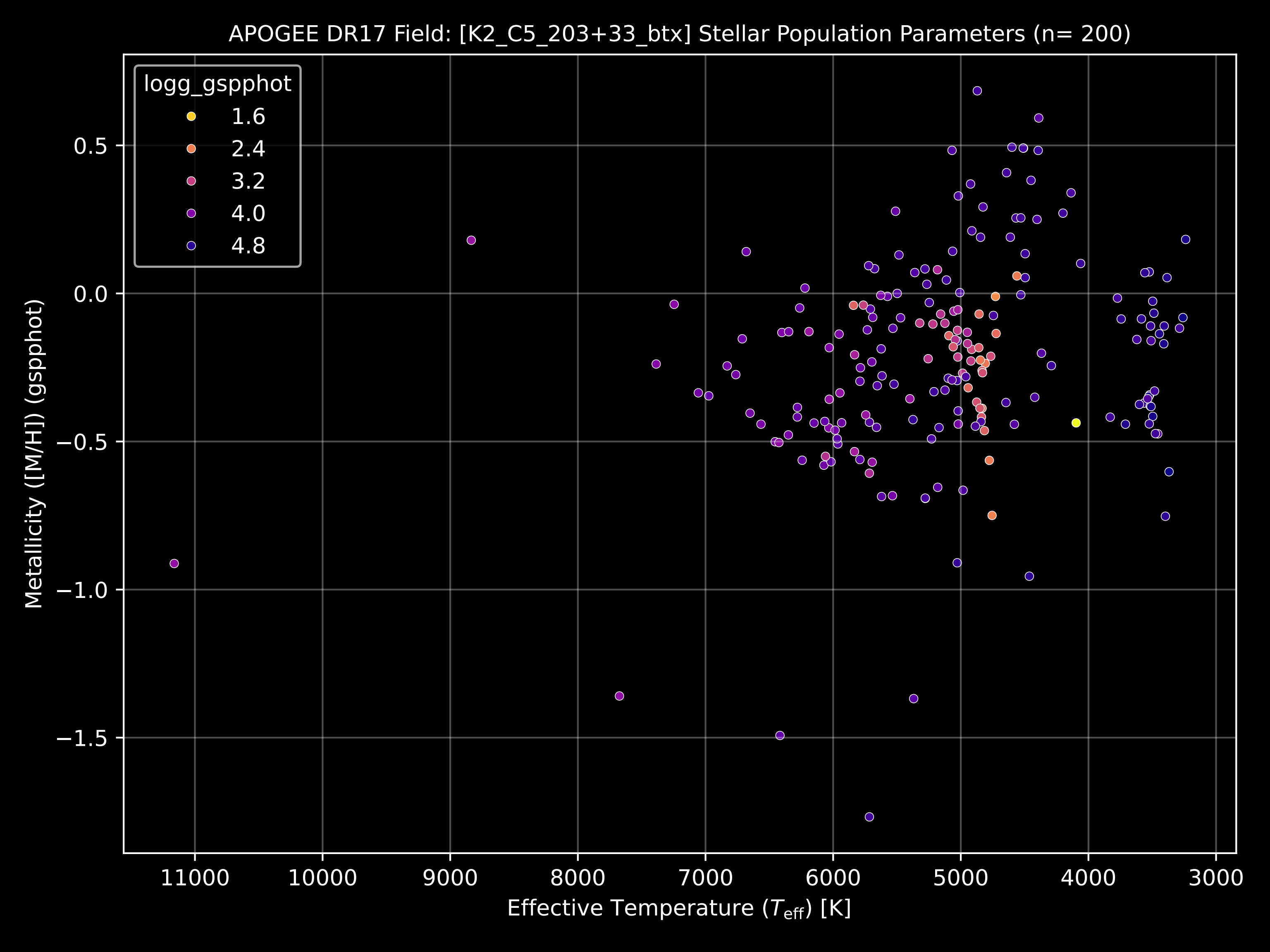

APOGEE DR17 Field: [K2_C5_203+33_btx]
Count plot of Star Type
Count
0
20
40
60
80
100
120
main_sequence
subgiants
turnoff
red_giants
red_clump
Star Type


APOGEE DR17 Field: [K2_C5_203+33_btx
[M/H] (mh_gspphot) (n= 198)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
−1.5
−1.0
−0.5
0.0
0.5

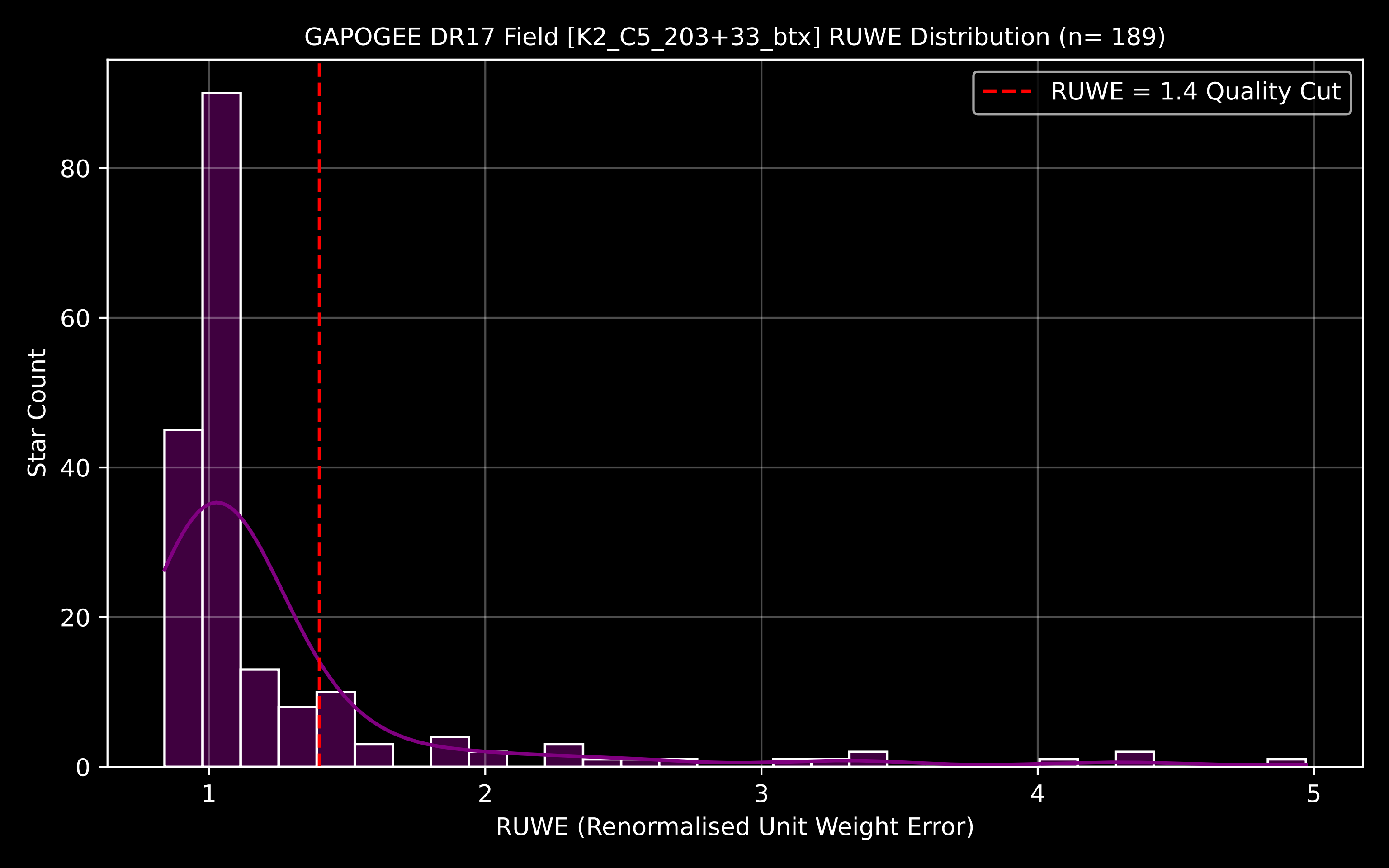

APOGEE DR17 Field: [K2_C5_203+33_btx] Proper Motion Diagram (n= 200)
bp_rp
0.6
1.2
1.8
2.4
3.0
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
20
0
−20
−40
−60
−80
−125
−100
−75
−50
−25
0
25
50



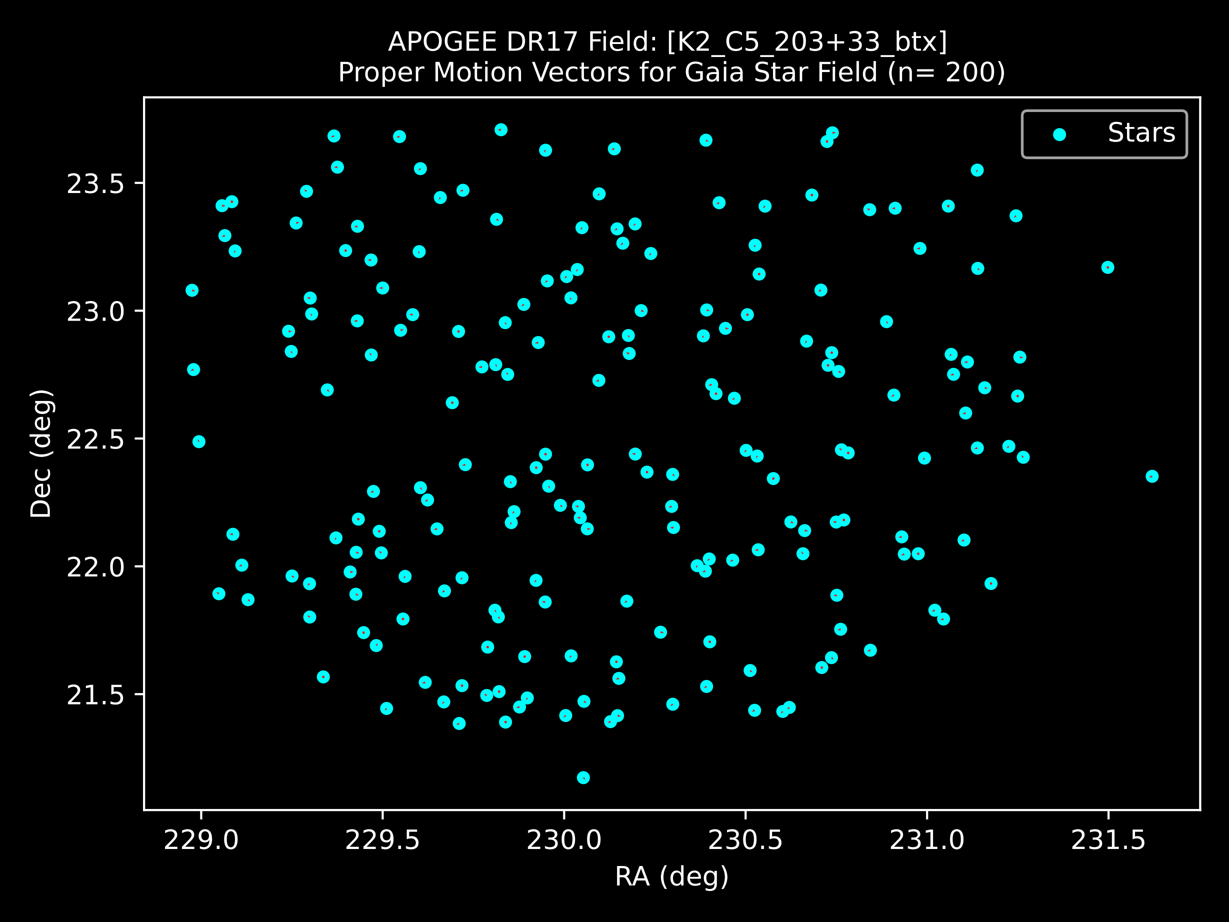
APOGEE DR17 Field: [K2_C5_203+33_btx] Parallax Quality (n= 200)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.0
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



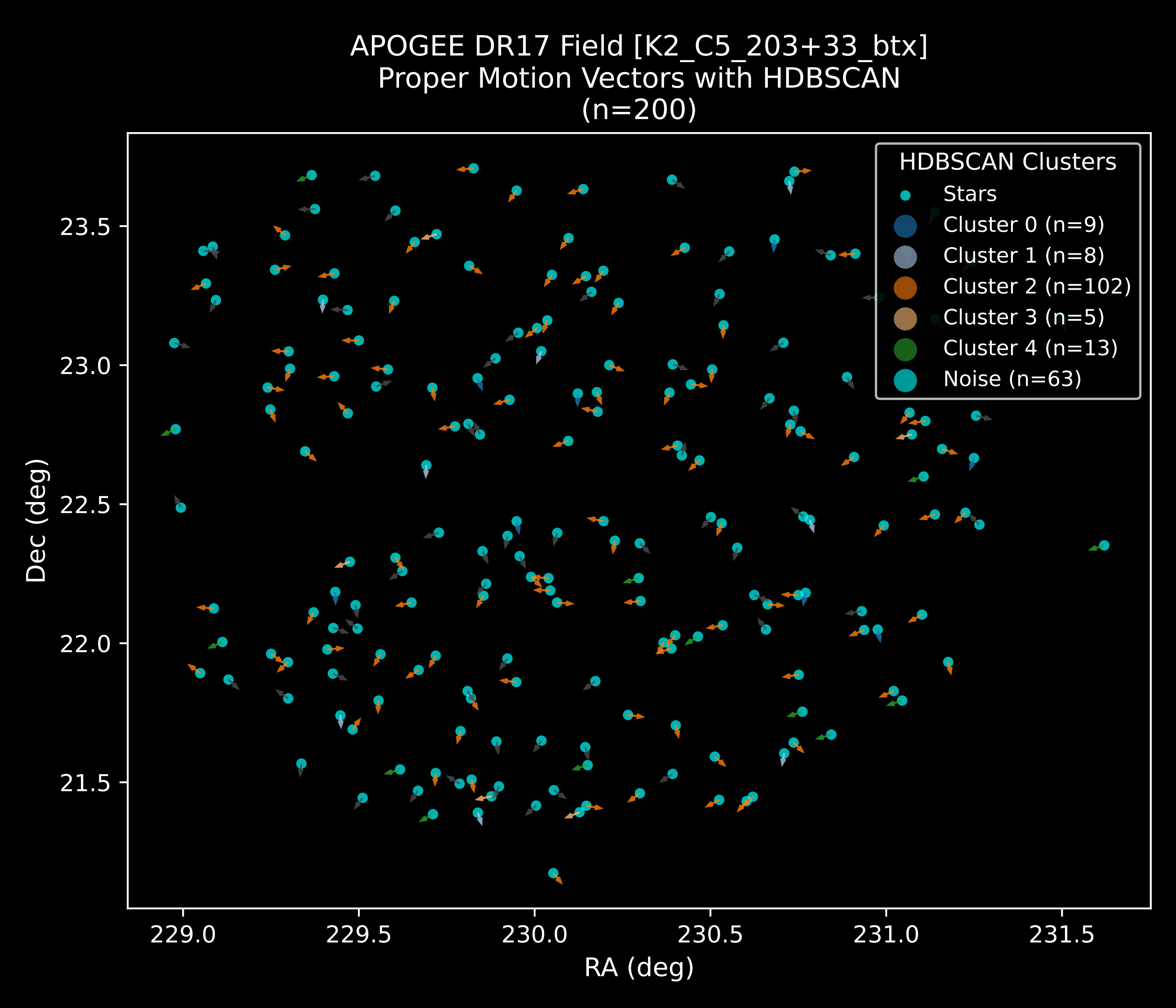
APOGEE DR17 Field: [K2_C5_203+33_btx] Stellar Population Parameters (n= 200)
logg_gspphot
1.6
2.4
3.2
4.0
4.8
0.5
0.0
−0.5
−1.0
−1.5
Metallicity ([M/H]) (gspphot)
11000
10000
9000
8000
7000
6000
5000
4000
3000
Effective Temperature (T_{eff}) [K]



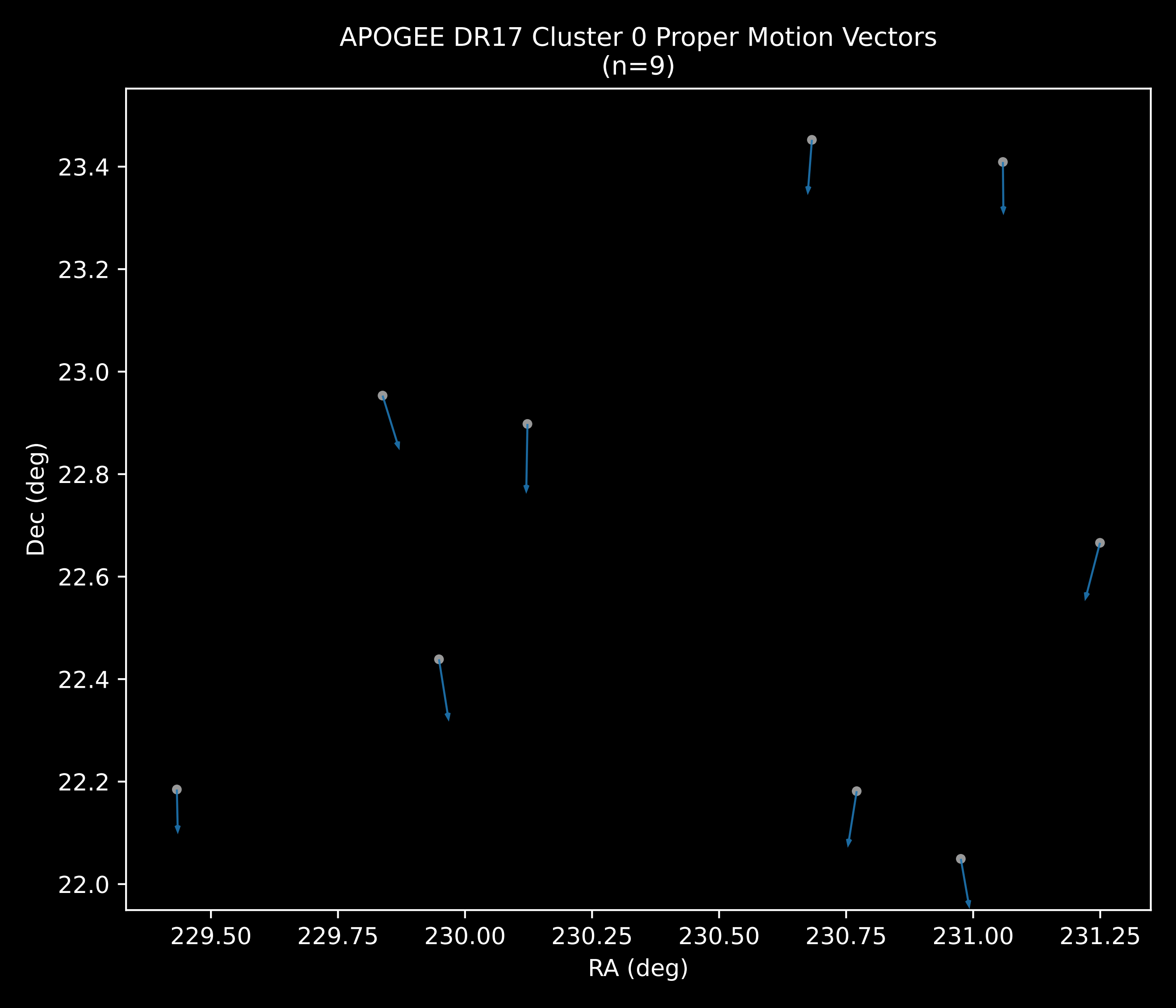
GAPOGEE DR17 Field [K2_C5_203+33_btx] RUWE Distribution (n= 189)
RUWE = 1.4 Quality Cut
Star Count
0
20
40
60
80
1
2
3
4
5
RUWE (Renormalised Unit Weight Error)

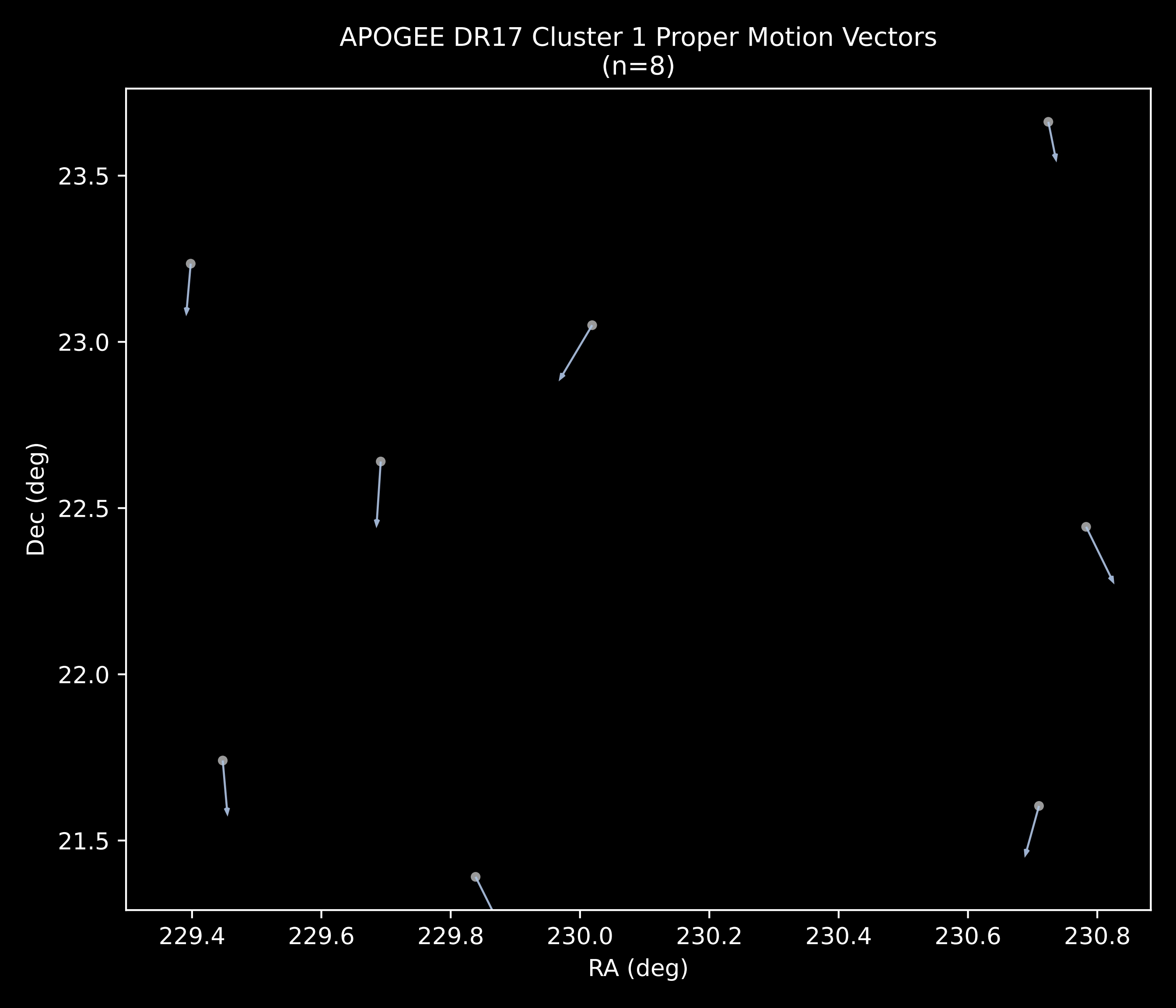


APOGEE DR17 Field: [K2_C5_203+33_btx]
Proper Motion Vectors for Gaia Star Field (n= 200)
Stars
Dec (deg)
RA (deg)
23.5
23.0
22.5
22.0
21.5
229.0
229.5
230.0
230.5
231.0
231.5

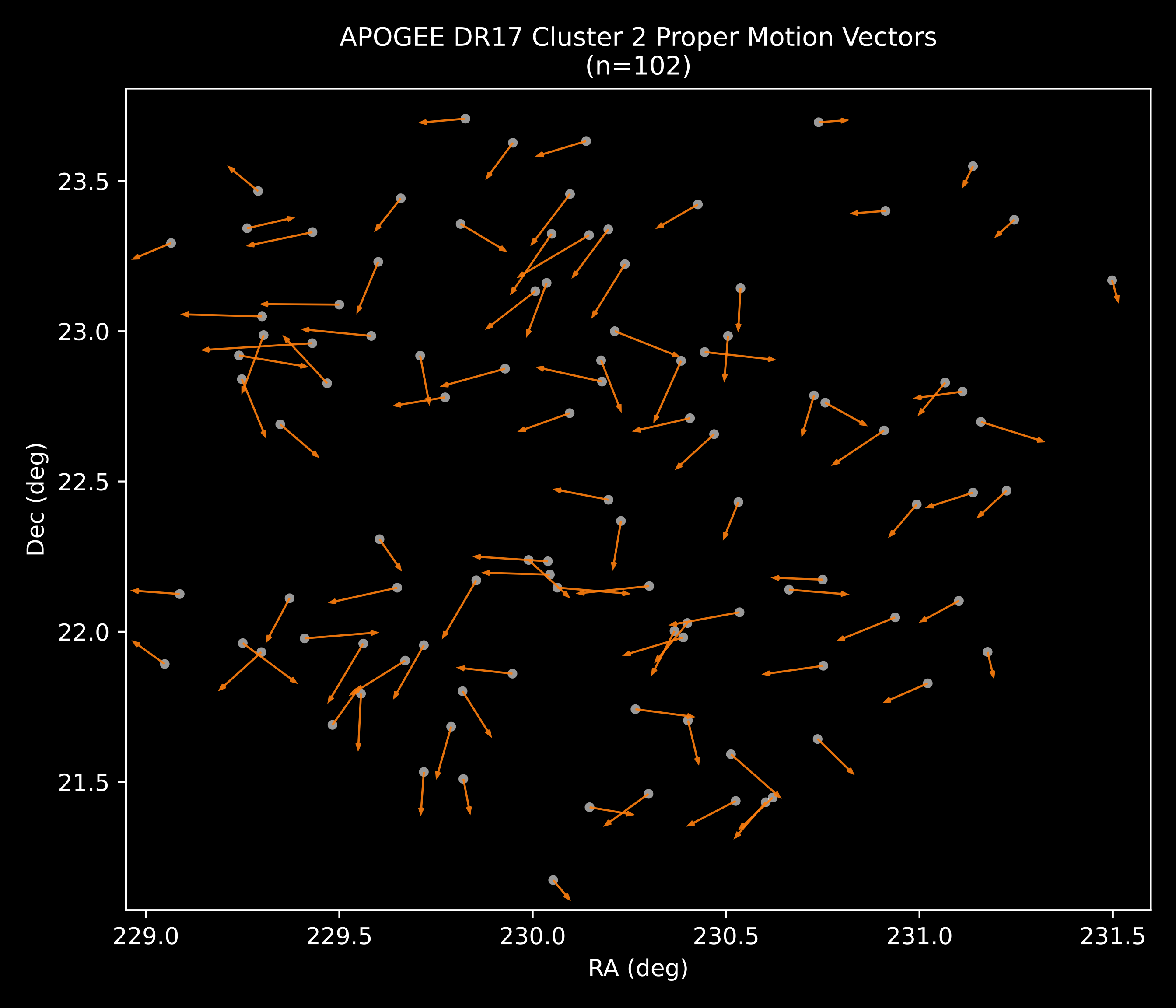


APOGEE DR17 Field [K2_C5_203+33_btx]
Proper Motion Vectors with HDBSCAN
(n=200)
HDBSCAN Clusters
Stars
Cluster 0 (n=9)
Cluster 1 (n=8)
Cluster 2 (n=102)
Cluster 3 (n=5)
Cluster 4 (n=13)
Noise (n=63)
Dec (deg)
RA (deg)
23.5
23.0
22.5
22.0
21.5
229.0
229.5
230.0
230.5
231.0
231.5

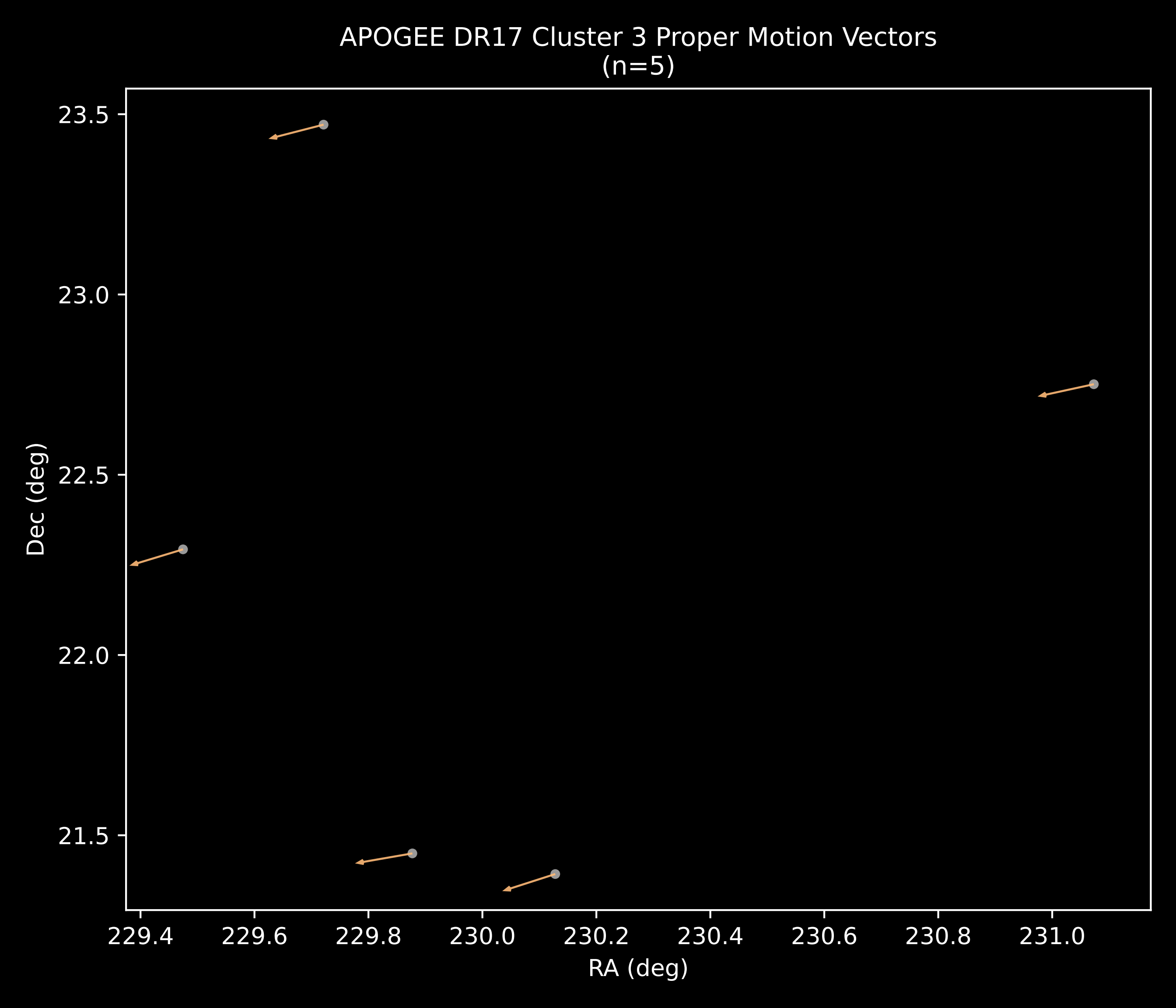

APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=9)
Dec (deg)
23.4
23.2
23.0
22.8
22.6
22.4
22.2
22.0
RA (deg)
229.50
229.75
230.00
230.25
230.50
230.75
231.00
231.25

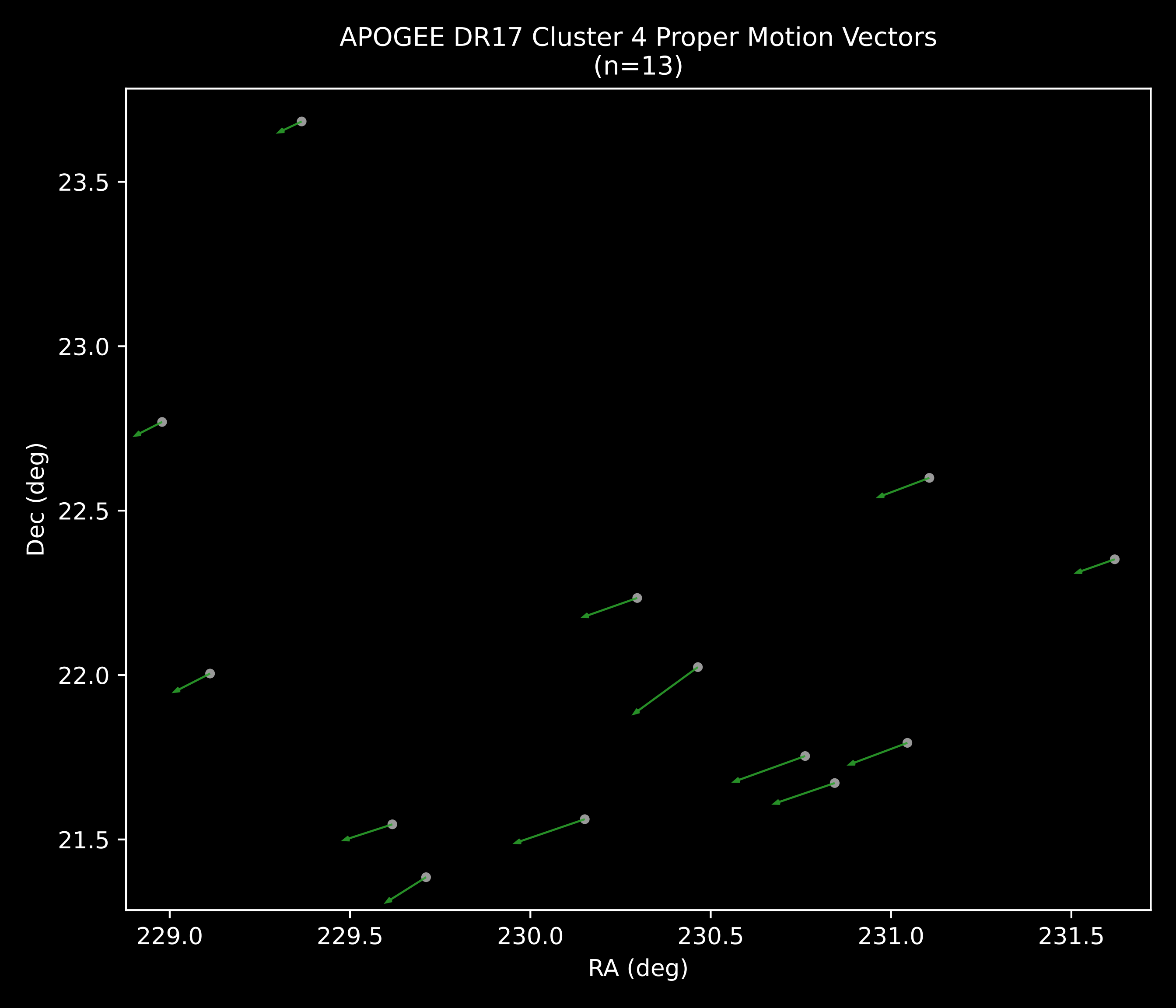


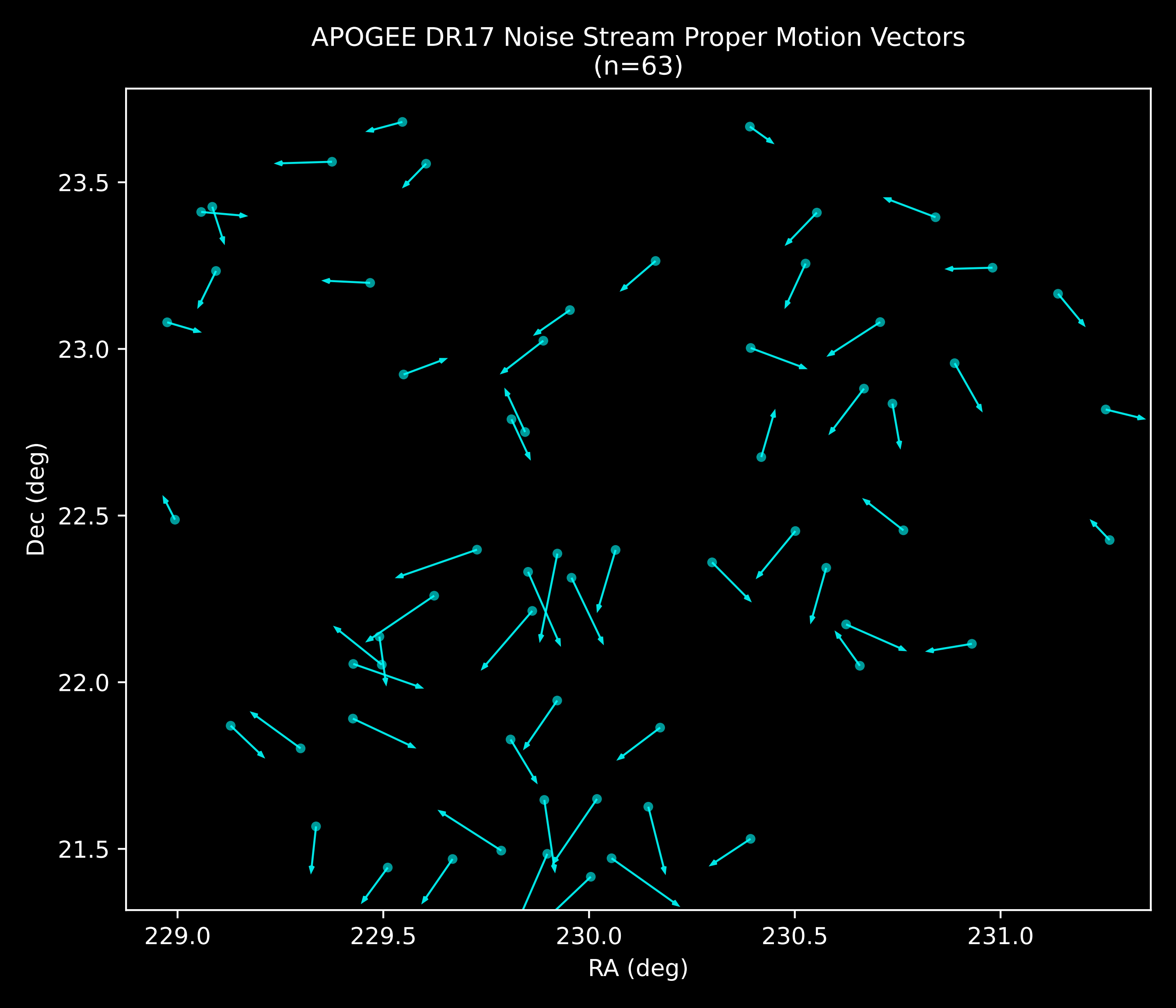
APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=8)
23.5
23.0
22.5
22.0
21.5
Dec (deg)
229.4
229.6
229.8
230.0
230.2
230.4
230.6
230.8
RA (deg)



APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=102)
Dec (deg)
23.5
23.0
22.5
22.0
21.5
RA (deg)
229.0
229.5
230.0
230.5
231.0
231.5


APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=5)
Dec (deg)
RA (deg)
23.5
23.0
22.5
22.0
21.5
229.4
229.6
229.8
230.0
230.2
230.4
230.6
230.8
231.0


APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=13)
23.5
23.0
22.5
22.0
21.5
Dec (deg)
229.0
229.5
230.0
230.5
231.0
231.5
RA (deg)


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=63)
Dec (deg)
RA (deg)
23.5
23.0
22.5
22.0
21.5
229.0
229.5
230.0
230.5
231.0